EXPERIMENT NO. 01

Title : Introduction of CAD software and to study and practice basic draw commands exists in the Auto CAD software.

Introduction of CAD :- Computer aided design involves the use of computer to create design drawing and product models computer aided design usually associated with interactive computer graphics. Computer aided design systems are powerful tools and are used in the mechanical design and geometric modelling of products and components.

Different CAD software :-

Different CAD software are :

1) Auto CAD.
2) Fusion 360
3) Catia.
4) Solidworks.
5) Creo parametric 3D modelling etc.

Introduction of Auto CAD : Auto CAD is most widely used software developed by auto desk. Auto CAD is a drafting package in almost all engineering branches. Auto CAD is one of the most popular CAD packages. It is a general purpose computer aided design we can draw geometrical entries like plan, section and elevation of a building, machine.

Basic Draw Commands in AutoCAD:

i) Line
ii) Poly line
iii) Construction line
iv) Polygon
v) Rectangle
vi) Arc
vii) Circle
viii) Ellipse

Theory of Draw commands:-

Line: The LINE commands is used to draw a line between two specified points. First enter the LINE command or shortcut L in the command area. Mouse cross hair displays to specify the first point of the line and next point of the line or enter the distance of the line. Press ESC key at the end of the line.

Polyline: Polyline command is used to draw a series of lines continuously at a time. Pl or PLINE is the keyboard shortcut for polyline.

Rectangle: Rectangle command is used to draw a rectangular general geometry in the drawings. REC is the keyboard shortcut for rectangle. First, enter the REC command in the command area. A cross hair will be displays to specify the first corner points of the rectangle and input the dimensions of the rectangle as other corner point.

Circle: Circles are created with the CIRCLE command. There are several different ways you can define a circle. Those are defining centre point and radius or diameter of the circle. In the 2 point method. Define the circle with points on either end of the circle diameter In the 3 points method. Define the circle with 3 nanocollineer points. In the Tangent, Tangent and Radius method define the circle by specifying two other objects that are tangent to the circle and the radius of the circle.

Arc: ARC command is used draw partial circles in the Auto CAD drawings. There are seven methods avaiable to draw an arc in the Auto CAD.

Those are:

i) Specifying 3 points on the arc
ii) Specifying Starting point, centre and end point.
iii) Specifying Starting point, centre and included angle.

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iv) Specifying starting point, center and length of the cord.
v) Specifying starting point, end point and radius.
vi) Specifying starting point, end point and included angle.
vii) Specifying starting point, end point and direction of start.

i) Polygon: The polygone command is used to draw a polygon with 3 to 1024 sides by with definite radius or by entering side length of the polygon.

) Ellipse: The ellipse command is used to draw ellipses by giving starting and ending angles.

Advantages :

1) We can make accurate and precise drawings like plan, section and elevation of a building, machine.

2) Improved engineering productivity.

3) Reduced engineering personal requirement.

4. Drawing modification or eraser to intake.

5. Drawings prepared in the software can be started safely.

Application of CAD:-

① Preparation of ~~Ani~~ Architectural drawings in the Auto CAD.

② Preparation of maps of an area

③ Preparation of Building drawings.

⑤ Preparation of Products design.

⑥ Preparation of circuit layouts for water distribution and sewer systems.

⑦ Design of Automobile parts, heat, exchangers, machine parts like drives, gears, screws etc.

Result:- Study of introduction to CAD, types of CAD software and practice of all basic commands using Auto CAD software.